FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Common Poisoning

Tutorial | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Aliku

Most childhood poisoning in Uganda is accidental ingestion by toddlers of paraffin, pesticides, household chemicals and adult medicines stored within reach; management is ABC and glucose first, then deciding whether decontamination helps or harms (never induce vomiting; no charcoal for paraffin, corrosives, iron or alcohol), then specific antidotes such as atropine, acetylcysteine, deferoxamine and antivenom.

Must know before the ward round

1. Order of care: ABC + glucose (treat as hypoglycaemia if you cannot measure in a drowsy child) → identify the poison (bring the container) → decontaminate only if it helps → antidote → supportive care → prevention counselling.
2. Never induce vomiting (no salt water, no fingers in the throat, no ipecac). Never give charcoal or do lavage after paraffin/petrol or corrosives.
3. Activated charcoal: 1 g/kg (UCG 2023: 0.5–1 g/kg; WHO: ≤1 year 1 g/kg, 1–12 years 25–50 g) within 1–2 hours, only if the airway is safe. Does NOT bind iron, lithium, alcohols, hydrocarbons, corrosives.
4. Paraffin (kerosene) is the commonest childhood poisoning in Uganda: the danger is aspiration pneumonitis. Observe at least 6 hours; oxygen if SpO_2 low; no emesis, lavage, charcoal, routine steroids or routine antibiotics.
5. Organophosphates: SLUDGE + killer Bs (bradycardia, bronchorrhoea, bronchospasm) + pinpoint pupils. Remove clothes and wash skin (staff wear gloves). Atropine 0.02 mg/kg IV/IM (WHO; UCG 0.05 mg/kg) doubled every 5 min until the chest is clear, HR normal, skin dry.
6. Pralidoxime 25–50 mg/kg IV over 30 min (max 2 g), then 10–20 mg/kg/h, for organophosphates with muscle weakness; useful only within about 24–48 h.
7. Paracetamol: toxic ≥150 mg/kg (UCG: 200 mg/kg if <6 years); acetylcysteine IV 150 mg/kg over 1 h, 50 mg/kg over 4 h, 100 mg/kg over 16 h; nearly 100% protective if started within 8 hours. Treat if 4-h level is above the treatment line (UK 100 mg/L).
8. Iron: work out ELEMENTAL iron (ferrous sulphate 200 mg tablet ≈ 60–65 mg). <20 mg/kg mild; >60 mg/kg serious. Charcoal useless. Deferoxamine 15 mg/kg/h IV (max 80 mg/kg/24 h, usually ≤24 h) for shock, acidosis, coma or severe symptoms.
9. Alcohol in small children causes hypoglycaemia, hypothermia and coma: give 10% dextrose 5 mL/kg, warm, protect the airway. Crude waragi may contain methanol.
10. Snakebite: immobilise, no tourniquet/cutting/sucking; 20-minute whole-blood clotting test; antivenom for systemic signs or rapidly spreading swelling. Children get the SAME antivenom dose as adults. Have adrenaline 0.01 mL/kg of 1:1000 IM ready.
11. Caustics (caustic soda, bleach, battery acid): no emesis, no neutralisation, no charcoal; airway first; nil by mouth if drooling/stridor; endoscopy within 12–24 h if symptomatic; watch for oesophageal stricture at 2–6 weeks.
12. Prevention: keep medicines and chemicals locked up and in original containers, NEVER store paraffin or pesticides in soda or water bottles.

1 Definition
2 Epidemiology
3 Aetiology and Risk Factors
4 Pathophysiology

5 Classification / Staging
6 Clinical Features
7 Differential Diagnoses
8 Clinical Workup
9 Management
10 Complications
11 Prognosis
12 Prevention
13 Special Populations
14 Key Points
15 References

1. Definition

Poisoning is the harmful effect of a substance swallowed, inhaled, injected or absorbed through the skin, in a dose that causes harm. In children it is usually **accidental** (a toddler exploring at 1–4 years, a dosing error by a caregiver); in adolescents it is often **intentional** (self-harm) or recreational (alcohol, drugs). **Envenoming** (snakebite) is managed on the same principles.

Term	Meaning
Poisoning	Exposure to a substance in a dose that causes harm
Toxidrome	A cluster of signs typical of a class of poison (e.g. cholinergic, anticholinergic, opioid, sympathomimetic)
Decontamination	Reducing absorption: skin/eye washing, activated charcoal, gastric lavage, whole-bowel irrigation
Antidote	A drug that reverses or blocks the poison (e.g. atropine, acetylcysteine, deferoxamine, naloxone, antivenom)
Enhanced elimination	Increasing removal of an absorbed poison (multiple-dose charcoal, urinary alkalinisation, dialysis)
"One pill can kill"	Drugs that are dangerous to a toddler in a single adult tablet: sulphonylureas, calcium-channel blockers, tricyclic antidepressants, opioids (including codeine), clonidine, chloroquine/quinine, methyl salicylate oil, camphor

Most poisoned children recover with good **supportive care**. Few need an antidote, and **many traditional "first aid" measures cause more harm than the poison** (salt water to induce vomiting, milk forced into a drowsy child, paraffin "to neutralise").

> **CLINICAL PEARL**
> The one thing never to forget: **ABC and glucose first, then think before you decontaminate.** Vomiting must never be induced, and charcoal or lavage after paraffin or corrosives can kill.

2. Epidemiology

- Poisoning causes a meaningful share of paediatric emergency admissions, peaking at **1–4 years** (mobile, curious, put everything in the mouth) with a second peak in **adolescence** (intentional).
- **Paraffin** used for lamps and stoves is the classic childhood poison in Ugandan homes: stored in **soda or water bottles** and placed on the floor.
- **Organophosphates** (e.g. chlorpyrifos, dimethoate, diazinon in dips and sprays) and **carbamates** are widely sold in unlabelled containers; children are exposed by ingestion, contaminated food, or skin contact (sprayed clothing, bedding, or treatment for lice and scabies with farm chemicals).
- **Caustic soda** (sodium hydroxide) used in home soap-making is a well-recognised cause of **oesophageal strictures** in Ugandan children, seen months later with dysphagia and wasting.
- **Snakebite**: most bites are by non-venomous snakes, and over half of bites by venomous species are "dry" (UCG 2023). Puff adders (cytotoxic), cobras and mambas (neurotoxic), and the boomslang and vine snake

(haemotoxic) are the important ones. Antivenom is scarce and expensive and is mostly held at regional and national referral hospitals.

- **Herbal medicines** are often given to children (enemas, oral "cleansing" mixtures) and can cause vomiting, diarrhoea, dehydration, liver and kidney injury.

> **UGANDA CONTEXT**
> Practical realities: **activated charcoal** is in the essential medicines list but is often out of stock at lower levels; **atropine** is usually available (it is used in theatre), **pralidoxime** much less so; **acetylcysteine** and **deferoxamine** are mainly at regional and national referral hospitals; **paracetamol levels** are rarely measurable outside Mulago and private labs. Polyvalent **antivenom** is held at some RRHs and Mulago. Decide early whether to **refer**, and phone ahead.

3. Aetiology and Risk Factors

Common poisons in Ugandan children: **paraffin (kerosene)** kept in soda bottles, **organophosphate and carbamate pesticides** used on farms and against fleas, ticks and bedbugs, **household chemicals** (caustic soda for soap-making, bleach, toilet cleaners, battery acid), **adult medicines** (paracetamol, iron, antimalarials, diabetes tablets, antiepileptics, ARVs), **alcohol** (sachets, local brew), **herbal medicines**, **rat poison** (warfarin-type), **carbon monoxide** (charcoal stove in a closed room) and **snakebite**.

Risk factors

Factor	Why
Age 1–4 years	Mobile, curious, mouthing
Unsafe storage: on the floor, in soda or water bottles, unlabelled containers	Looks like a drink; no warning
Poor supervision, siblings as carers	Unwitnessed ingestion
Adult medicines in the home (ANC iron and folate, diabetes, antiepileptics, ARVs)	"One pill can kill" drugs; iron is common after pregnancy
Farming households	Pesticides, dips and sprays; contaminated clothing
Home soap-making	Caustic soda
Dosing errors by caregivers (paracetamol syrups, codeine syrups, herbal mixtures)	Repeated supratherapeutic ingestion
Adolescence: low mood, school or relationship stress, abuse	Intentional overdose, often multiple drugs (assume paracetamol)
Rural living, walking barefoot at night, sleeping on the floor	Snakebite
Previous poisoning	Marker of an unsafe environment

4. Pathophysiology

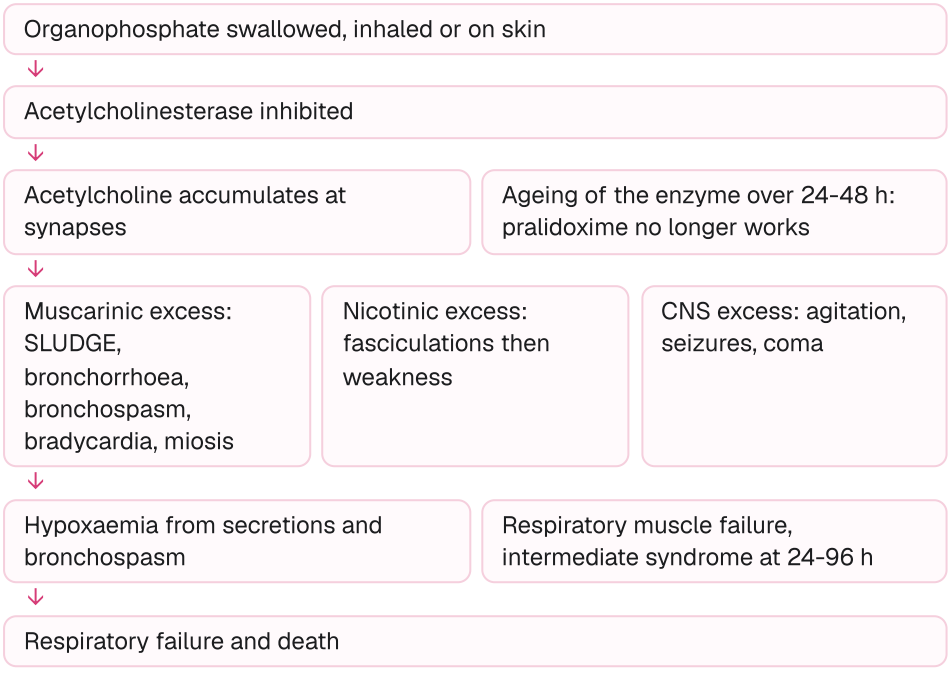
Why decontamination can harm

- **Induced vomiting** does not remove much of the poison, delays charcoal, and risks **aspiration**, especially of hydrocarbons, which have low viscosity and surface tension and spread across the lungs causing **chemical pneumonitis**. Salt-water emetics can cause fatal **hypernatraemia**.
- **Corrosives** burn on the way down; vomiting burns the oesophagus a **second time** and can cause perforation.

- **Activated charcoal** adsorbs many drugs in the gut, but it does **not bind** small ionised or metallic molecules: **iron, lithium, potassium, alcohols (ethanol, methanol), hydrocarbons, acids and alkalis.** It is risky if the child is drowsy (aspiration) and is most useful **within 1 hour** (up to 2 hours per UCG; longer for slow-release drugs).

Mechanisms of the key poisons

Poison	Mechanism	Consequence
Paraffin/kerosene	Aspiration of low-viscosity hydrocarbon → destroys surfactant, inflames alveoli; CNS depression if absorbed	Cough, tachypnoea, hypoxaemia, fever (chemical), pneumatocoeles; drowsiness, seizures
Organophosphates	**Irreversibly** inhibit acetylcholinesterase ("ageing" over 24–48 h) → acetylcholine excess at muscarinic, nicotinic and CNS synapses	Cholinergic crisis, then weakness, respiratory failure; intermediate syndrome at 24–96 h
Carbamates	**Reversibly** inhibit acetylcholinesterase	Similar but shorter-lived; less CNS toxicity
Paracetamol	Excess saturates glucuronidation/sulphation → more NAPQI → glutathione depleted → hepatocyte necrosis (zone 3)	Liver failure at 72–96 h; acetylcysteine replaces glutathione
Iron	Direct corrosion of GI mucosa; free iron in blood → mitochondrial toxicity, vasodilatation, capillary leak	Vomiting/haematemesis, shock, metabolic acidosis, liver failure, later pyloric stricture
Ethanol	CNS depressant; **blocks hepatic gluconeogenesis**	Coma, hypoglycaemia, hypothermia (children are very sensitive)
Methanol	Metabolised to formic acid	Severe acidosis, blindness, death after a latent period of 12–24 h
Caustics	Alkalis → liquefactive necrosis (deep, oesophagus); acids → coagulative necrosis (stomach)	Burns, perforation, stricture, later oesophageal cancer risk
Snake venom	Cytotoxic (local necrosis), neurotoxic (post-synaptic or pre-synaptic blockade → paralysis), haemotoxic (consumptive coagulopathy)	Swelling, paralysis, bleeding, shock, AKI



Feature links

- Muscarinic acetylcholine excess → salivation, lacrimation, urination, defecation, vomiting, sweating, bronchorrhoea, bronchospasm, bradycardia, pinpoint pupils.
- Nicotinic excess at the neuromuscular junction → fasciculations → weakness → respiratory failure; same mechanism later → intermediate syndrome.
- Atropine blocks muscarinic receptors only → dries the chest but does not reverse weakness (pralidoxime does, before ageing).
- Too much atropine → anticholinergic picture: hot, delirious, ileus, urinary retention.
- Hydrocarbon aspiration → surfactant loss and alveolitis → cough, tachypnoea, crackles, low SpO_2, chemical fever in the first 24–48 h, pneumatocoeles.
- Hydrocarbon absorption → CNS depression → drowsiness, seizures.
- NAPQI accumulation → zone 3 necrosis → raised ALT at 24 h, liver failure at 72–96 h (jaundice, INR, hypoglycaemia, encephalopathy).
- Iron corrosion of the gut → vomiting, haematemesis, later pyloric stricture; free iron → capillary leak and mitochondrial poisoning → shock, metabolic acidosis, liver failure.
- Ethanol blocking gluconeogenesis → hypoglycaemia → coma, seizures; vasodilatation → hypothermia.
- Methanol → formic acid → raised anion gap acidosis, deep breathing, blindness after a latent period.
- Salicylate → respiratory centre stimulation plus metabolic acidosis → deep, fast breathing, vomiting, tinnitus, fever.
- Alkali liquefactive necrosis → drooling, stridor, refusal to swallow → perforation → healing by fibrosis → stricture at 2–6 weeks.
- Cytotoxic venom → local swelling, blistering, necrosis, compartment syndrome; haemotoxic venom → consumptive coagulopathy → non-clotting blood, bleeding gums; neurotoxic venom → ptosis, bulbar palsy, respiratory paralysis.
- Opioids → mu-receptor agonism → pinpoint pupils, slow breathing, coma.
- Anticholinergics → hot, dry, flushed, dilated pupils, delirium, urinary retention, tachycardia.

5. Classification / Staging

Common toxidromes

Toxidrome	Signs	Examples
Cholinergic	**SLUDGE** (Salivation, Lacrimation, Urination, Defecation, GI upset, Emesis) + **killer Bs** (Bradycardia, Bronchorrhoea, Bronchospasm), **miosis**, sweating, fasciculations, weakness, seizures	Organophosphates, carbamates
Anticholinergic	"Hot as a hare, dry as a bone, red as a beet, blind as a bat, mad as a hatter": fever, dry skin, flushing, **dilated pupils**, delirium, urinary retention, tachycardia	Antihistamines, Datura (jimsonweed) seeds, tricyclics
Opioid	**Pinpoint pupils**, coma, slow breathing	Codeine cough syrups, morphine, tramadol
Sympathomimetic	Agitation, tachycardia, hypertension, hyperthermia, dilated pupils, **sweating**	Amphetamines, khat (older children), salbutamol overdose
Sedative-hypnotic	Drowsiness, ataxia, slurred speech, respiratory depression, normal pupils	Phenobarbital, diazepam, alcohol

Toxidrome	Signs	Examples
Salicylate	Vomiting, tinnitus, **deep fast breathing**, fever, mixed acid–base disturbance	Aspirin, methyl salicylate (oil of wintergreen)

Severity by dose

Poison	Threshold
Paracetamol (single acute)	**≥150 mg/kg** toxic (UCG: ≥200 mg/kg if <6 years); repeated supratherapeutic ≥150 mg/kg/24 h
Iron (elemental)	**<20 mg/kg**: usually GI upset only; **20–60 mg/kg**: possible systemic toxicity; **>60 mg/kg**: serious toxicity likely
Serum iron (4–6 h)	**>60 µmol/L** moderate; **>90 µmol/L (500 µg/dL)** severe

Snakebite by venom type: cytotoxic (puff adder: local swelling and necrosis), neurotoxic (cobras, mambas: paralysis), haemotoxic (boomslang, vine snake: bleeding); **dry bite** = no envenoming (no signs at 6–8 h).

Caustic burns are graded at endoscopy (Zargar): grade 1–2a → feed early; severe (2b and above) → NG/gastrostomy feeding and high stricture risk.

6. Clinical Features

History

The "**5 Ws**": **Who** (age, weight), **What** (the substance; bring the container, tablet strip, plant, the snake if killed), **When** (time of exposure), **How much** (maximum possible: count missing tablets, measure the bottle), **Why/how** (accident, error, self-harm, abuse).

Question	Why it matters
Exact substance, brand and strength (container, label, tablet strength)	Toxicity depends on the dose per kg; e.g. ferrous sulphate 200 mg = 60–65 mg elemental iron; look up the active ingredient
Maximum amount possibly taken	Assume the worst: all missing tablets, a mouthful = about 5–10 mL in a toddler, 15 mL in an older child
Time since ingestion	Decides whether charcoal helps (<1–2 h), when to take paracetamol levels (4 h), and when symptoms are expected
Symptoms since: vomiting, cough/choking, drowsiness, convulsions, drooling, pain on swallowing, vomiting blood	Aspiration, CNS toxicity, corrosive injury, iron
"First aid" already given: salt water, milk, herbs, induced vomiting, oil	Complications and aspiration risk
Other children exposed?	Siblings may also be poisoned: examine them too
Weight, past medical history, HIV status, current medicines in the home	Dosing; chronic illness affects risk
Immunisation (tetanus)	Snakebite and burns need tetanus prophylaxis
Social history: supervision, storage, previous poisoning, adolescent mood and stressors	Prevention, **safeguarding** (abuse or neglect), **self-harm assessment**

Examination

In a poisoned child the **ETAT/ABCDE assessment comes first** (section 9). Once the child is stable, complete the examination in the order of Hutchison's Clinical Methods (24th edition), adapted for children.

Introduction, consent and preparation

- Introduce yourself, confirm the child's identity and the carer's relationship; explain and gain consent (assent from an older child).
- **Protect yourself**: gloves and apron for organophosphates and caustics; **hand hygiene**.
- **Position and exposure**: remove contaminated clothing (bag it), keep the child warm; drowsy children in the recovery position.

General inspection (end of the bed)

- Conscious level (**AVPU/GCS**), distress, seizures, posture.
- **Smell**: paraffin, alcohol, "garlic" or chemical smell (organophosphates), acetone.
- Colour (cyanosis, pallor), sweating or dry flushed skin, drooling, secretions.
- **Vitals**: RR, **SpO_2**, HR, BP, CRT, **temperature** (hypothermia with alcohol; fever with anticholinergics, salicylate, chemical pneumonitis), **glucose**.
- **Weight** (for mg/kg calculations) and nutrition.

Hands, face and neck

- **Hands and skin**: sweating (cholinergic, sympathomimetic) vs dry and flushed (anticholinergic); fasciculations; **fang marks, swelling, bruising, blistering** (snakebite; mark and time the edge of the swelling).
- **Eyes and pupils**: pinpoint (opioids, organophosphates), dilated (anticholinergics, sympathomimetics); **ptosis** (neurotoxic snakebite, OP weakness); lacrimation; chemical or venom in the eyes.
- **Mouth**: burns, ulcers, **drooling** (caustics); salivation (OP); bleeding gums (haemotoxic venom).
- **Neck**: **stridor**, hoarseness (caustic airway oedema); neck flexor weakness (intermediate syndrome).

Systems

- **Respiratory**: work of breathing; deep acidotic breathing (salicylate, methanol, iron); crackles and wheeze (hydrocarbon pneumonitis, organophosphate bronchorrhoea).
- **Cardiovascular**: bradycardia (OP), tachycardia (anticholinergics, salbutamol), arrhythmia (tricyclics), shock (iron, envenoming).
- **Abdomen**: tenderness (iron, caustics), hepatomegaly or liver tenderness (late paracetamol), bowel sounds (absent in atropine toxicity), **bladder** (retention in anticholinergic or atropine toxicity).
- **Neurological**: fasciculations, weakness, ptosis, bulbar palsy (organophosphate, neurotoxic snakebite), ataxia, seizures.

Completing the examination

Examine any other exposed children; reassess ABC; thank the family, **hand hygiene**, summarise and document.

Document these positives	And these negatives
Time of exposure and time seen	"No burns to lips, mouth or pharynx"
RR, SpO_2, HR, BP, temperature, GCS/AVPU, pupils	"Chest clear; no crackles or wheeze"
Toxidrome features present	"No fasciculations or weakness"
For snakebite: extent of swelling (mark and time it), clotting test	"No ptosis; no bleeding from gums or venepuncture sites"

Red flags

RED FLAG

- Reduced consciousness, seizures, respiratory distress or SpO_2 <94%.
- **Drooling, stridor, inability to swallow** after a caustic ingestion (airway and oesophageal burns).
- **Bronchorrhoea and bradycardia** (organophosphate): atropine now.
- **Shock, haematemesis, metabolic acidosis** after iron.
- Paracetamol ingestion **≥150 mg/kg**, or unknown amount with a delay over 8 hours.
- **Ptosis, difficulty swallowing, weakness** after a snakebite (neurotoxicity can progress to respiratory paralysis); **bleeding gums or non-clotting blood** (haemotoxicity).
- A "**one pill can kill**" drug (sulphonylurea, calcium-channel blocker, tricyclic, opioid).
- **Intentional ingestion** in an adolescent (risk of repeat; possible multiple drugs).
- Call your senior immediately for: coma, seizures, respiratory failure, shock, cholinergic crisis, caustic ingestion with drooling/stridor, iron >60 mg/kg or symptomatic, paracetamol ≥150 mg/kg, any envenomed snakebite, or intentional ingestion.

Thinking like a clinician

- **Treat the patient, not the poison.** ABC, glucose and seizure control first; most children survive with supportive care.
- **Assume the worst-case dose** and calculate mg/kg.
- **A well child now may not stay well**: paracetamol, iron, sulphonylureas, modified-release drugs and methanol have **delayed toxicity**. Observe for the right time.
- **A child "who drank paraffin" and is coughing** has aspirated; a child who swallowed paraffin without coughing or choking is unlikely to develop pneumonitis but still needs 6 hours of observation.
- **Organophosphate poisoning can look like pneumonia or gastroenteritis** (secretions, crackles, diarrhoea, vomiting): look at the **pupils** and the **heart rate**, and smell the child.
- **Unexplained coma in a toddler**: hypoglycaemia (alcohol, sulphonylurea), opioids (codeine syrup), sedatives, organophosphates, carbon monoxide (charcoal stove in a closed room), and of course cerebral malaria and meningitis.
- **The atropine target is a dry chest**, not big pupils or a fast heart.
- **Children get the adult antivenom dose**: the dose neutralises the venom injected, not the body weight.
- **Common student mistakes**: inducing vomiting; giving charcoal after paraffin or to a drowsy child; giving iron poisoning charcoal; under-dosing atropine (OP needs large, escalating doses); stopping observation too early; forgetting siblings and prevention counselling; missing intentional overdose in adolescents.

EXAM FAVOURITE

Viva favourite: **"A toddler swallowed paraffin 30 minutes ago and is coughing. What will you do?"** ABC and SpO_2; **do not induce vomiting, no lavage, no charcoal**; remove soaked clothes and wash skin; oxygen if SpO_2 <94%/<90%; observe at least 6 hours; chest X-ray if symptomatic; no routine antibiotics or steroids; counsel on storage.

7. Differential Diagnoses

Poisoning mimics common illnesses, and common illnesses can be blamed on poisoning.

Differential	Distinguishing clinical feature	Distinguishing investigation
Coma: cerebral malaria, meningitis, hypoglycaemia, head injury, DKA (vs alcohol, opioids, sedatives, OP, CO, sulphonylurea)	Fever, neck stiffness, trauma, Kussmaul breathing vs smell, pupils, history of access	Glucose, mRDT, LP when safe
Seizures: febrile convulsion, malaria, meningitis, epilepsy (vs OP, tricyclics, isoniazid, hypoglycaemic agents, kerosene, camphor)	Fever; known epilepsy vs toxidrome	Glucose, mRDT
Respiratory distress: pneumonia, asthma, bronchiolitis, foreign body (vs hydrocarbon aspiration, OP bronchorrhoea, salicylate acidotic breathing)	Paraffin smell, pinpoint pupils and bradycardia, deep breathing without chest signs	SpO_2, CXR, pupils
Vomiting and diarrhoea: gastroenteritis (vs iron, OP, herbal mixtures, mushrooms, food poisoning)	Haematemesis (iron), cholinergic signs, herbal history	History, glucose, Hb, abdominal X-ray (iron tablets)
Bleeding: sepsis/DIC, ITP, leukaemia, haemophilia (vs snakebite, warfarin-type rat poison)	Bite, swelling; rat poison in the home	20WBCT, platelets, INR
Painful swollen limb after a bite: cellulitis, abscess, fracture, insect sting (vs cytotoxic snakebite)	Fang marks, rapid spread, blistering	Clotting test
Stridor and drooling: croup, epiglottitis, foreign body (vs caustic ingestion)	Mouth burns, history of caustic soda or bleach	Clinical; endoscopy

8. Clinical Workup

Diagnosis is usually from the history, the container and the toxidrome. Tests confirm severity and guide antidotes; never delay ABC, glucose or an indicated antidote for a test.

Bedside

Test	When / how	Expected finding	How it changes management	Available at
Glucose	Every drowsy child; alcohol, sulphonylureas, salicylate, iron, severe poisoning	Low	10% dextrose 5 mL/kg	All levels with a glucometer
SpO_2	Hydrocarbon, OP, opioid, neurotoxic snakebite	<94%	Oxygen; admission	HC IV and above
20-minute whole-blood clotting test (20WBCT)	Every snakebite on arrival and every 4–6 h: 2 mL blood in a clean, dry **glass** tube, undisturbed 20 min, then tipped once	**Not clotted = coagulopathy** (UCG 2023 describes observing after 30 minutes) Verify *UCG 20 vs 30 minute reading*	Antivenom; repeat 6 h after antivenom	All levels
Urine dipstick	Snakebite, salicylate	Blood (haemolysis, myoglobin), ketones	AKI care; fluids	All levels
ECG	Tricyclics, calcium-channel blockers, OP	Bradycardia, QT prolongation, broad QRS	Bicarbonate (tricyclics); monitoring	Hospitals

Test	When / how	Expected finding	How it changes management	Available at
Haemoglobin	Bleeding after snakebite or iron	Low	Transfusion	HC IV and above
Dose calculation	Every ingestion	mg/kg (paracetamol, elemental iron)	Decides treatment	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
LFTs (ALT/AST), INR, creatinine	Paracetamol, iron, herbal, snakebite	ALT >1000 IU/L, INR rising, AKI	Continue NAC; refer for liver/renal care	RRH/national referral
PT/INR, platelets, fibrinogen	Snakebite, rat poison	Prolonged, low	Antivenom; vitamin K for warfarin-type rat poison	Hospitals
Electrolytes, creatinine	Iron, salicylate, methanol, severe poisoning, snakebite	Acidosis, AKI, potassium shifts	Bicarbonate, fluids, dialysis	General hospital and above

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Chest X-ray	Hydrocarbon ingestion with symptoms, or at 6 h	Perihilar/basal infiltrates (may lag 2–8 h), pneumatocoeles	Oxygen, admission	Hospitals
Abdominal X-ray	Iron (tablets radio-opaque), button batteries	Tablets in stomach/intestine; battery position	Whole-bowel irrigation; urgent endoscopy for batteries in oesophagus	Hospitals
Chest/abdominal X-ray	Caustic ingestion with suspected perforation	Free air, mediastinal air	Surgery	Hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Serum paracetamol (4 h or later)	Paracetamol ingestion	Above the treatment line (UK **100 mg/L at 4 h**; US 150 mg/L)	Acetylcysteine	Mulago/private labs; often not available, so treat on dose
Serum iron (4–6 h)	Iron ingestion >20 mg/kg	>60 µmol/L moderate; >90 µmol/L severe	Deferoxamine	Rare
Blood gas	Iron, salicylate, methanol, severe poisoning	Metabolic acidosis, raised anion gap	Bicarbonate, antidote, dialysis	Mulago, some RRHs
Plasma cholinesterase / RBC acetylcholinesterase	Uncertain OP diagnosis	Low	Confirms diagnosis (does not guide initial treatment)	Rarely available
Upper GI endoscopy	Caustic ingestion with symptoms	Grade of burns (Zargar)	Feeding plan, stricture risk	Mulago and a few RRHs/private

Test	When	Expected / abnormal finding	How it changes management	Available at
Drug/alcohol screen	Unknown ingestion	Positive	Rarely changes acute care	Rarely available

Putting it together

Model one-line summary: "A 2-year-old boy, 12 kg, who swallowed an unknown amount of paraffin from a soda bottle 1 hour ago, now with cough, RR 52 and SpO_2 91% in air; **hydrocarbon aspiration pneumonitis**, no CNS toxicity."

Problem list (model case below)

1. Paraffin ingestion with aspiration pneumonitis, hypoxaemic.
2. Aspiration risk worsened by induced vomiting (salt water).
3. Low-grade fever: likely chemical, observe.
4. Unsafe storage at home: prevention counselling.

Model case presentation

> "Master T.O., a **22-month-old boy** from Nateete, **weight 11 kg** (weight-for-age Z-score −0.5, MUAC 14.0 cm), brought at 16:30 by his mother **45 minutes after drinking paraffin** from a soda bottle kept on the kitchen floor. He **coughed and choked immediately** and vomited once at home after the neighbour gave salt water. On arrival: alert, RR **54**, mild subcostal recession, SpO_2 **92% in air**, scattered crackles at the right base, temperature 37.9 °C, glucose 5.2 mmol/L. **Fully immunised for age** (child health card), **breastfeeding plus family foods**, **mother HIV-negative** this year; no previous poisoning. **Assessment: paraffin ingestion with aspiration pneumonitis (worsened by induced vomiting). Plan**: oxygen by nasal prongs to SpO_2 ≥94%, nil by mouth while distressed then cup feeds, maintenance IV fluids if not drinking, chest X-ray, observe RR and SpO_2 hourly, no charcoal, no antibiotics unless fever persists beyond 48 hours or he deteriorates, and **counsel on safe storage** (never in soda bottles) and on not inducing vomiting."

NOTE

Worked example: is this iron ingestion dangerous? A 3-year-old (14 kg) ate up to 10 tablets of ferrous sulphate 200 mg from her mother's ANC supply. Elemental iron per tablet ≈ 60–65 mg → up to **650 mg** ÷ 14 kg = **46 mg/kg**. This is a **potentially moderate–severe ingestion** (>40 mg/kg). She vomited twice and is now quiet. **Action**: admit, IV access, glucose, abdominal X-ray (count tablets), serum iron at 4–6 h if possible, blood gas; **no charcoal**; if she develops shock, acidosis, lethargy or repeated vomiting/haematemesis, start **deferoxamine 15 mg/kg/h** and refer.

NOTE

Worked example: paracetamol by dose. A 4-year-old (16 kg) drank most of a 100 mL bottle of paracetamol syrup 120 mg/5 mL 3 hours ago. Maximum dose = 100 ÷ 5 × 120 = **2,400 mg** ÷ 16 kg = **150 mg/kg**. UCG says treat ≥150 mg/kg (≥200 mg/kg if under 6), so this is borderline; with no level available and a full bottle missing, **start IV acetylcysteine** within 8 hours, check LFTs/INR at baseline and at the end of the infusion.

9. Management

Immediate / emergency

First 5 minutes (any poisoned child)

1. **Protect yourself** (gloves and apron for organophosphates, caustics).
2. **ETAT ABCD**: open and clear the airway (suction secretions), oxygen to SpO_2 ≥94% if emergency signs, IV access, treat shock (10–20 mL/kg crystalloid).
3. **Glucose**: check; if low or unmeasurable in a drowsy child, **10% dextrose 5 mL/kg**.
4. **Convulsions**: diazepam 0.5 mg/kg rectally or 0.2 mg/kg IV (max 10 mg).
5. **Decontaminate the skin/eyes** if exposed (remove clothes, wash with soap and water; irrigate eyes for 10–15 minutes).
6. **Life-saving antidote** if the toxidrome is obvious: **atropine** (cholinergic crisis), **naloxone** (opioid with respiratory depression), **adrenaline** (anaphylaxis to antivenom).
7. **Identify the poison**: container, label, smell, toxidrome; call senior.

General algorithm

1. ABC + glucose + seizures (ETAT); protect yourself.
2. Identify: what, how much (mg/kg), when, symptoms.
3. Decontaminate **only if benefit > risk**: skin and eyes always; charcoal 1 g/kg if <1–2 h, airway safe and a charcoal-binding poison; **no** emesis; **no** charcoal or lavage for hydrocarbons, corrosives, iron, alcohol.
4. Antidote if indicated (drug table).
5. Supportive care: O_2, fluids, glucose, warmth, observation for the right time.
6. Prevention counselling; safeguarding; psychosocial review if intentional.

Decontamination rules

Method	Use	Do NOT use
Remove clothes, wash skin with soap and water	OP, carbamates, hydrocarbons on skin, caustics on skin	Always do it; staff wear gloves
Eye irrigation 10–15 min with water/saline	Caustics, spitting-cobra venom, chemicals	Always do it
Activated charcoal	Most drugs within **1 hour** (UCG up to 2 h; longer for slow-release/anticholinergic)	**Drowsy or unprotected airway; hydrocarbons; corrosives; iron, lithium, alcohols; bowel obstruction**
Gastric lavage	Rarely: life-threatening ingestion <1 h, protected airway, experienced staff (24–28 Fr orogastric tube, 10 mL/kg saline aliquots)	Corrosives, hydrocarbons, drowsy child without intubation
Induced vomiting	**Never** (no salt water, no ipecac, no fingers)	Never
Whole-bowel irrigation (PEG solution)	Iron, slow-release drugs, packets (specialist)	Obstruction, perforation, unstable child
Water or milk sips	Caustic ingestion **only** if the child is alert and can swallow, immediately after exposure (WHO)	Drooling, stridor, vomiting, reduced consciousness

UGANDA CONTEXT

UCG 2023 still lists **gastric lavage** for some ingestions (e.g. organophosphates up to 6 hours, paracetamol within 2 hours); current toxicology practice (WHO Pocket Book, AACT/EAPCCT) restricts lavage to life-threatening ingestions within about 1 hour by experienced staff with a protected airway. Discuss with your senior before attempting it. UCG and WHO also differ on the **atropine starting dose** (UCG 0.05 mg/kg; WHO 0.02 mg/kg) and on **charcoal dosing** (UCG 0.5–1 g/kg; WHO ≤1 year 1 g/kg, 1–12 years 25–50 g).

Definitive

Paraffin (kerosene) and other hydrocarbons

1. ABC; remove soaked clothes and wash the skin.
2. **Do not induce vomiting, do not lavage, do not give charcoal, do not give milk or oil.**
3. **Oxygen** if SpO_2 <94% with distress (<90% if stable); position of comfort; nil by mouth if distressed.
4. **Observe at least 6 hours** with RR and SpO_2.
   - **Asymptomatic at 6 hours** (no cough, normal RR and SpO_2, normal chest X-ray if done): discharge with advice to return if cough or fast breathing develops; review in 24 h.
   - **Symptomatic**: admit; chest X-ray (changes may appear 2–8 h after ingestion); oxygen; IV fluids at maintenance if not drinking.
5. **Fever** in the first 24–48 h is usually chemical; **antibiotics** only if fever persists beyond 48 h, a new infiltrate develops, or the child deteriorates (secondary bacterial pneumonia).
6. **Steroids**: no benefit.
7. Severe cases (ARDS, pneumothorax) need CPAP or ventilation at national referral level.

Organophosphates and carbamates

1. **Protect staff** (gloves, apron). **Remove all clothes**, wash skin and hair with soap and water, irrigate eyes. Bag the clothes.
2. **Airway**: suction secretions; oxygen; recovery position; bag-mask ventilation if weak or apnoeic (respiratory failure is the killer).
3. **Atropine** early, without waiting for oxygen saturation to improve: **0.02 mg/kg IV** (UCG: 0.05 mg/kg), **doubling every 5 minutes** until **atropinisation**:
   - clear chest on auscultation (no wheeze or crackles from secretions)
   - heart rate normal for age (above about 80 in older children, higher in infants)
   - dry axillae
   - pupils no longer pinpoint
   - adequate blood pressure

   Then an **infusion** (10–20% of the total loading dose per hour), titrated hourly; reduce slowly over 24–48 h.
4. **Pralidoxime** 25–50 mg/kg IV over 30 min, then 10–20 mg/kg/h if muscle weakness/fasciculations or moderate–severe poisoning (UCG: only if not responding to adequate atropine; most useful within 24 h).
5. **Seizures**: diazepam.
6. **Charcoal** 1 g/kg if ingested within 1–2 h and the airway is safe (most pesticides are in hydrocarbon solvents, so aspiration is a risk; **never induce vomiting**).
7. **Monitor for days**: **intermediate syndrome** (24–96 h): weakness of neck flexors, proximal limbs, cranial nerves and **respiratory muscles**, even after the cholinergic crisis has settled; delayed neuropathy at 2–3 weeks.

DRUG DOSING

Atropine toxicity vs under-atropinisation: if the child is hot, delirious, with absent bowel sounds and a full bladder, **stop** the infusion for 30 minutes and restart at 70–80% of the rate. If secretions and bradycardia return, **increase**. Dilated pupils and tachycardia alone are not the targets; a **dry chest** is.

Paracetamol

1. Calculate the dose in mg/kg (assume the worst). **Single acute ingestion ≥150 mg/kg** (UCG: ≥200 mg/kg if <6 years; liquid ingestions in toddlers rarely reach toxicity) or **unknown amount** → treat.
2. **Charcoal 1 g/kg** if within 1 hour (UCG: within 2 hours; repeated doses listed but not routine in WHO/UK practice).
3. **Paracetamol level at 4 hours** (or as soon as possible after 4 h) and LFTs/INR/creatinine; plot on the nomogram (UK treatment line **100 mg/L at 4 h**; US 150 mg/L).
4. **Acetylcysteine IV** (21-hour, three-bag regimen) if above the line, if level unavailable and dose toxic, if **>8 hours** since ingestion with a toxic dose (start immediately, don't wait for the level), if **staggered** ingestion, or if LFTs are abnormal. **Nearly 100% protective when started within 8 hours.**
5. At the end of NAC: repeat ALT/INR; continue NAC if ALT is rising or INR >1.3; refer liver failure (encephalopathy, INR >3, acidosis, hypoglycaemia, AKI).
6. **Repeated supratherapeutic ingestion** (≥150 mg/kg/24 h, Medscape): treat if ALT is raised or the level is detectable.

Iron

1. Calculate **elemental iron** in mg/kg: ferrous sulphate 200 mg ≈ 60–65 mg; ferrous fumarate 200 mg ≈ 65 mg; ferrous gluconate 300 mg ≈ 35 mg. **Prenatal iron + folate tablets** (common in Ugandan homes) contain about 60 mg elemental iron.
2. **<20 mg/kg**: usually only GI upset; observe 6 hours at home or clinic if asymptomatic. **20–60 mg/kg**: possible systemic toxicity; admit and observe. **>60 mg/kg**: serious toxicity likely.
3. **No charcoal.** Abdominal X-ray for tablets; **whole-bowel irrigation** (PEG) if many tablets are seen (specialist); lavage is of limited value.
4. **Supportive**: IV fluids for hypovolaemia (GI losses, capillary leak), glucose, correct acidosis.
5. **Deferoxamine 15 mg/kg/h IV** if shock, metabolic acidosis, altered consciousness, persistent vomiting/haematemesis, or serum iron >90 µmol/L; reduce after 4–6 h, total ≤80 mg/kg/24 h (WHO), usually stop within 24 h (lung toxicity with prolonged infusion).
6. A child **asymptomatic at 6 hours** is very unlikely to develop severe toxicity.
7. Watch for **hepatic failure** (2–4 days) and **pyloric/duodenal stricture** (2–6 weeks).

Alcohol (ethanol) and methanol

1. ABC, recovery position, protect from aspiration.
2. **Glucose**: check; **10% dextrose 5 mL/kg**, then a dextrose infusion until eating (UCG; children are very prone to hypoglycaemia).
3. **Warm** the child (hypothermia).
4. Observe until fully awake; exclude head injury and other drugs.
5. **Methanol** (suspect crude/illegal brew, several people affected, visual symptoms, severe acidosis after a latent period): refer urgently; UCG lists **oral ethanol 40% (e.g. waragi) 1.5–2 mL/kg loading** then 0.3 mL/kg/h as the antidote where fomepizole is unavailable, plus bicarbonate and dialysis.
6. Social history: who gave the alcohol, supervision, adolescent drinking; counselling.

Snakebite

1. **First aid** (UCG 2023): reassure, keep still, lie on the side, remove rings and tight items, **immobilise the limb with a splint**; for likely **neurotoxic** bites a firm crepe bandage from the fingers/toes upwards (WHO Pocket Book). **No tourniquet, no cutting, no sucking, no herbs, no attempt to kill the snake** (but bring it if already dead).
2. **Hospital**: ABC; IV access in the **unbitten** limb; **20WBCT** on arrival and every 4–6 hours; mark the edge of the swelling with the time; analgesia with **paracetamol** (avoid NSAIDs); **tetanus** prophylaxis; avoid IM injections.
3. **Observe at least 12–24 hours**; no signs at 6–8 hours suggests a dry bite (UCG).
4. **Antivenom indications** (UCG/WHO): **systemic envenoming** (neurotoxicity: ptosis, bulbar palsy, respiratory weakness; coagulopathy: non-clotting blood, bleeding; shock) or **rapidly spreading local swelling** (hand/foot within 1 h, elbow/knee within 3 h, groin/chest at any time, more than half the limb, or significant head/neck swelling).
5. **Giving antivenom**: polyvalent (Africa) antivenom; **children get the same dose as adults** (venom load is the same); dilute in 2–3 volumes of 0.9% saline and give IV over 1 hour, slowly at first; **adrenaline 1:1000 drawn up** (0.01 mL/kg IM) with chlorphenamine and hydrocortisone at hand. Repeat after 6 hours if blood still does not clot, or after 1–2 hours if neurotoxicity or bleeding worsens.
6. **Neurotoxicity**: airway and ventilation (bag-mask by relays if no ventilator), **atropine followed by neostigmine** (anticholinesterase trial) can reverse post-synaptic (cobra) paralysis.
7. **Local necrosis**: surgical review, debridement once demarcated, fasciotomy only if compartment syndrome is proven; antibiotics only if infected.
8. **Spitting cobra venom in the eyes**: irrigate with plenty of water, cover; ophthalmology review.

Caustics (corrosives)

1. **Airway first**: stridor, drooling, hoarseness → senior and anaesthetist; intubation may be difficult (oedema).
2. **No emesis, no neutralisation** (acids with alkalis or vice versa produce heat), **no charcoal, no lavage**.
3. If alert, swallowing and **seen within minutes**: small sips of water or milk (WHO); otherwise **nil by mouth**.
4. IV fluids, analgesia (morphine 0.05–0.1 mg/kg IV), PPI/H2 blocker (e.g. omeprazole 1 mg/kg daily).
5. **Endoscopy within 12–24 hours** (not after 48–72 h, when the wall is weakest) if symptomatic (drooling, vomiting, stridor, refusal to swallow, burns in the mouth); grade the burns; mild (grade 1–2a) → feed early; severe → NG/gastrostomy feeding.
6. Chest/abdominal X-ray if perforation suspected (mediastinitis, peritonitis).
7. **Steroids** are not routinely recommended; antibiotics only for perforation or infection.
8. **Follow up** for **oesophageal stricture** (dysphagia 2–6 weeks later) → serial dilatation at Mulago or refer; growth monitoring; nutrition.
9. **Button batteries**: an emergency of their own (oesophageal burns within 2 hours): X-ray and urgent endoscopic removal.

Intentional ingestion (adolescents)

Ask about co-ingestants (**assume paracetamol** and check a level where possible), monitor, and arrange a **psychosocial/psychiatric assessment before discharge**; involve the family and ensure safety at home (remove access to medicines and pesticides).

Drug table

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Activated charcoal (charcoal-binding drug, <1–2 h, safe airway)	**1 g/kg** (UCG: 0.5–1 g/kg; WHO: ≤1 y 1 g/kg; 1–12 y 25–50 g); max 50 g/dose	Oral or NG, mixed with 8–10 volumes of water	Once; multiple-dose 0.5–1 g/kg every 4 h for phenobarbital, carbamazepine, quinine, salicylates	As indicated	Not for hydrocarbons, corrosives, iron, lithium, alcohols, drowsy child
Atropine (OP/carbamate cholinergic crisis)	**0.02 mg/kg** (WHO Pocket Book; max 2 mg per dose); UCG 2023: **0.05 mg/kg**	IV (IM if no IV)	**Double every 5 minutes** until atropinised; then infusion of 10–20% of the total loading dose per hour (UCG 0.05 mg/kg/h)	Reduce slowly over 24–48 h	Toxicity: delirium, fever, ileus, urinary retention
Pralidoxime (OP with weakness/fasciculations, moderate–severe)	**25–50 mg/kg** (max 2 g)	IV over 30 min	Then **10–20 mg/kg/h** infusion (or repeat bolus 6–12-hourly)	Until recovery	Most useful within 24–48 h (before ageing); not routinely for carbamates
Diazepam (seizures)	**0.2 mg/kg IV** or **0.5 mg/kg rectal**; max 10 mg/dose	IV/rectal	Repeat once after 10 min	Acute	Respiratory depression
Acetylcysteine (NAC) IV (paracetamol)	**150 mg/kg over 1 h**, then **50 mg/kg over 4 h**, then **100 mg/kg over 16 h** (total 300 mg/kg in 21 h); max 15 g, 5 g, 10 g for the three bags (UCG)	IV in 5% dextrose (for <20 kg: 3 mL/kg, 7 mL/kg, 14 mL/kg respectively, WHO)	Continuous	21 h; continue if late, ALT rising or INR raised	Anaphylactoid reactions: slow or pause, antihistamine, restart
Acetylcysteine oral (if IV unavailable)	**140 mg/kg** load, then **70 mg/kg**	Oral/NG	4-hourly	× 17 doses (72 h)	Vomiting common; give antiemetic
Deferoxamine (iron with shock, acidosis, coma, severe symptoms, or high serum iron)	**15 mg/kg/h**; max **80 mg/kg/24 h** (WHO), max 6 g/day	IV infusion in 0.9% saline or 5% dextrose	Continuous; reduce after 4–6 h (WHO)	Usually ≤24 h	"Vin rosé" urine; hypotension if too fast; lung toxicity if prolonged; contraindicated in anuria (UCG)
10% dextrose (alcohol, sulphonylurea, salicylate, hypoglycaemia)	**5 mL/kg**	IV	Then 5–10% dextrose infusion	Until eating	Recheck glucose at 30 min
Naloxone (opioid with respiratory depression: codeine syrups, tramadol, morphine)	**0.01 mg/kg** (10 µg/kg), increase to **0.1 mg/kg** if no response (UCG)	IV/IM	Repeat every 2–3 min; infusion if needed	Until opioid effect wears off	Aim to restore breathing, not full consciousness; short acting

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Vitamin K (phytomenadione) (warfarin-type rat poison with bleeding or raised INR)	**0.3 mg/kg** (max 10 mg; UCG adult 5 mg)	Slow IV	Daily	Long-acting rodenticides may need weeks	FFP if active bleeding
Polyvalent antivenom (envenoming)	**Same dose as adults** (per package insert)	IV over 1 h in 2–3 volumes of 0.9% saline	Repeat after 6 h if blood still non-clotting; sooner if worsening neurotoxicity or bleeding	Until clotting restored	Have adrenaline drawn up
Adrenaline 1:1000 (antivenom reaction/anaphylaxis)	**0.01 mL/kg** (0.15 mL for small children, WHO); max 0.5 mL	IM thigh	Every 5 min if needed	Acute	
Neostigmine (post-synaptic neurotoxic snakebite, after atropine)	Per specialist protocol Verify *paediatric dose not given in the source notes*	IV	Trial	Acute	Give atropine first
Sodium bicarbonate (salicylate urinary alkalinisation; tricyclic cardiotoxicity)	**1 mmol/kg** (UCG bolus 1–2 mEq/kg)	IV	Then infusion to keep urine pH 7.5–8	Until toxicity resolves	Monitor potassium
Ethanol 40% (e.g. waragi) (methanol, where fomepizole unavailable)	**1.5–2 mL/kg** loading, then **0.3 mL/kg/h** (UCG)	Oral/NG	Continuous	Until methanol cleared	Hypoglycaemia, sedation; refer
Morphine (caustic burns pain)	**0.05–0.1 mg/kg**	IV	As needed	Short	Respiratory depression
Omeprazole (caustic injury)	**1 mg/kg**	Oral/IV	Daily	Per endoscopy findings	
Paracetamol (snakebite pain)	15 mg/kg	Oral	6-hourly	Short	Avoid NSAIDs in snakebite

Referral criteria

Level	What to do	Refer up when
Home/community (first aid)	Take the child and container to a health facility; do not induce vomiting; wash skin; for caustics, sips of water if alert; immobilise snakebitten limbs	Always seek care

Level	What to do	Refer up when
HC II/III	ABC, glucose, diazepam, skin decontamination, charcoal if indicated and safe, atropine for OP crisis, oxygen if available	**All symptomatic children, paracetamol/iron overdoses and snakebites**, with a note
HC IV / General Hospital	As above plus observation for hydrocarbons, IV fluids, atropine infusion, 20WBCT, NAC if stocked	Need for antivenom, NAC, deferoxamine, pralidoxime, endoscopy, ventilation
RRH / National Referral (Mulago)	NAC, deferoxamine, pralidoxime, antivenom, endoscopy, ICU/ventilation, dialysis	Liver failure, ARDS, oesophageal dilatation

Phone ahead before transfer and send the container or tablet strip with the child.

Monitoring and follow-up

- **Observation times**: paraffin **≥6 h**; iron **6 h** (asymptomatic at 6 h → very unlikely to develop severe toxicity); snakebite **12–24 h** (20WBCT every 4–6 h; mark swelling); OP for **days** (intermediate syndrome 24–96 h); paracetamol until NAC complete and ALT/INR checked; alcohol until fully awake and glucose stable.
- **OP**: hourly chest, HR, axillae, pupils, BP, bowel sounds and bladder while titrating atropine; neck flexor strength and breathing for intermediate syndrome.
- **Hydrocarbon**: RR and SpO_2 hourly; temperature (fever beyond 48 h → secondary infection).
- **Paracetamol**: ALT/INR at the end of NAC; continue if ALT rising or INR >1.3.
- **Caustics**: follow up for **dysphagia at 2–6 weeks** (stricture); growth.
- **Iron**: pyloric stricture at 2–6 weeks.
- **Snakebite**: wound, function, contractures.
- **Before discharge**: prevention counselling for every family; safeguarding concerns addressed; psychosocial assessment after intentional ingestion.

10. Complications

Disease-related

Poison	Complications
Paraffin	Aspiration pneumonitis, ARDS, pneumatocoeles, pneumothorax, secondary bacterial pneumonia, seizures
Organophosphates	Respiratory failure, aspiration, intermediate syndrome (24–96 h), delayed polyneuropathy (2–3 weeks)
Paracetamol	Acute liver failure (72–96 h), AKI, hypoglycaemia, death
Iron	Shock, metabolic acidosis, liver failure, gastric outlet/pyloric stricture (2–6 weeks)
Alcohol	Hypoglycaemia, hypothermia, aspiration, seizures
Methanol	Severe acidosis, blindness, death
Snakebite	Local necrosis, compartment syndrome, amputation, bleeding (including intracranial), respiratory paralysis, AKI, contractures
Caustics	Airway oedema, perforation, mediastinitis, **oesophageal stricture**, malnutrition, later oesophageal carcinoma

Treatment-related

- **Induced vomiting** (traditional first aid): aspiration; salt-water emetics → fatal hypernatraemia; second burn in caustic ingestion.

- **Charcoal**: aspiration in a drowsy child.
- **Lavage**: aspiration, oesophageal injury.
- **Atropine toxicity**: hyperthermia, delirium, ileus, urinary retention.
- **Acetylcysteine**: anaphylactoid reactions.
- **Deferoxamine**: hypotension if infused too fast; lung toxicity if prolonged beyond 24 h.
- **Antivenom reactions**: anaphylaxis (adrenaline drawn up), later serum sickness.
- **Naloxone**: short action, so opioid effect can return.

11. Prognosis

- **Most poisoned children recover with good supportive care**; few need an antidote.
- **Paraffin**: most recover; asymptomatic children at 6 hours can go home.
- **Paracetamol**: acetylcysteine is **nearly 100% protective when started within 8 hours**; delay increases the risk of liver failure.
- **Iron**: a child asymptomatic at 6 hours is very unlikely to develop severe toxicity.
- **Organophosphates**: outcome depends on early airway care and adequate atropine; pralidoxime only helps before ageing (24–48 h).
- **Caustics**: long-term morbidity from oesophageal stricture (dysphagia, wasting) and later carcinoma risk.
- **Snakebite**: most bites are dry or by non-venomous snakes; envenomed children risk necrosis, amputation and disability.
- **Adolescent self-harm** carries a risk of repeat attempts; follow-up with psychosocial support matters as much as the antidote.

12. Prevention

Counsel every family:

- **Store** medicines, paraffin, pesticides and chemicals **locked away and out of reach** (above 1.5 m), in **original labelled containers**; **never in soda or water bottles**.
- Buy pesticides only in labelled containers; never decant into food containers; keep children away during spraying.
- **Child-resistant containers** and blister packs.
- Do not call medicines "sweets"; supervise children when adults take medicines.
- Dispose of old medicines safely.
- **First aid**: never induce vomiting; take the child and the container to the health facility.
- **Snakebite**: clear bush around houses, wear shoes at night, use a torch, sleep under a net on a raised bed.
- **Carbon monoxide**: never use a charcoal stove in a closed room.
- **Adolescents**: ask about mood, school, relationships, abuse; involve counsellors.

13. Special Populations

- **Toddlers (1–4 years)**: "one pill can kill" drugs; very prone to **hypoglycaemia** with alcohol; liquid paracetamol ingestions rarely reach toxicity.
- **Adolescents**: intentional, often multiple drugs; assume paracetamol; psychosocial assessment, safeguarding and family involvement before discharge.
- **Suspected abuse or neglect** (repeated poisoning, unexplained ingestion in an infant who cannot walk): safeguarding referral.

- **Snakebite in children**: the **same antivenom dose as adults**, because the venom load is the same; higher venom-to-body-mass ratio means faster systemic effects.
- **Pregnant adolescents** with poisoning: treat the mother; most antidotes (NAC, antivenom, atropine) are still indicated.

14. Key Points

High-yield points

- ABC and glucose first, then think before you decontaminate.
- **Never induce vomiting**; no charcoal or lavage after paraffin or corrosives.
- Charcoal does not bind iron, lithium, alcohols, hydrocarbons, acids and alkalis.
- Paraffin: observe at least 6 hours; no routine antibiotics or steroids.
- OP: atropine doubling every 5 minutes until the **chest is dry**.
- Paracetamol ≥150 mg/kg: NAC, nearly 100% protective within 8 hours.
- Iron: calculate **elemental** iron; deferoxamine for systemic toxicity.
- Snakebite: immobilise, 20WBCT, **adult antivenom dose** with adrenaline ready.
- Caustics: airway, no neutralisation, endoscopy within 12–24 h, watch for stricture.
- Counsel every family: never store paraffin or pesticides in soda bottles.

Numbers & doses

Item	Value
Activated charcoal	**1 g/kg** (UCG 0.5–1 g/kg; max 50 g), within 1–2 h, safe airway
Paraffin observation	**≥6 hours**
Atropine (OP)	**0.02 mg/kg** IV (WHO; UCG 0.05 mg/kg), **double every 5 min** until chest dry
Pralidoxime	**25–50 mg/kg** IV over 30 min (max 2 g), then **10–20 mg/kg/h**
Paracetamol toxic dose	**≥150 mg/kg** (UCG ≥200 mg/kg if <6 y)
Paracetamol treatment line	**100 mg/L at 4 h** (UK)
NAC IV	**150 mg/kg/1 h → 50 mg/kg/4 h → 100 mg/kg/16 h**
NAC oral	140 mg/kg, then 70 mg/kg 4-hourly × 17
Iron toxicity	**<20 mg/kg** mild; **>60 mg/kg** severe (elemental)
Ferrous sulphate 200 mg	**≈60–65 mg elemental iron**
Deferoxamine	**15 mg/kg/h IV**, max 80 mg/kg/24 h, usually ≤24 h
Naloxone	**0.01 mg/kg**, up to 0.1 mg/kg
10% dextrose	**5 mL/kg**
Diazepam	**0.5 mg/kg rectal** or **0.2 mg/kg IV** (max 10 mg)
20WBCT	Non-clotting at 20 min = coagulopathy
Antivenom	**Adult dose** for children; repeat after 6 h if still non-clotting
Adrenaline (antivenom reaction)	**0.01 mL/kg of 1:1000 IM**, max 0.5 mL
Caustic endoscopy	Within **12–24 h** (avoid 48–72 h)
OP intermediate syndrome	**24–96 h**

Exam traps

- **"Give salt water or milk to make the child vomit."** Wrong: never induce vomiting.
- **"Activated charcoal for paraffin."** Wrong: aspiration risk and no binding.
- **"Charcoal for iron tablets."** Wrong: iron is not adsorbed; use X-ray, bowel irrigation, deferoxamine.
- **"Atropine 0.6 mg once for OP."** Wrong: escalating, **doubling** doses until the chest is dry; children may need large totals.
- **"Paracetamol level at 1 hour to decide NAC."** Wrong: levels before **4 hours** are uninterpretable; if >8 h, start NAC first.
- **"Smaller antivenom dose for a small child."** Wrong: **same dose** as adults.
- **"Neutralise the alkali with vinegar."** Wrong: the reaction generates heat and worsens burns.
- **"Routine antibiotics and steroids for kerosene pneumonitis."** Wrong: supportive care; antibiotics only for secondary infection.

Questions seniors ask

Viva questions

1. *Why never induce vomiting?* It removes little poison, delays definitive care and causes aspiration (hydrocarbons) and re-exposure of the oesophagus (corrosives); salt-water emetics can cause fatal hypernatraemia.
2. *Name five poisons charcoal does not bind.* Iron, lithium, alcohols (ethanol, methanol), hydrocarbons (paraffin, petrol), acids and alkalis.
3. *What are the endpoints of atropinisation?* Clear chest, heart rate normal for age, dry axillae, pupils no longer pinpoint, adequate BP.
4. *When is paracetamol toxic in a child?* ≥150 mg/kg (UCG: ≥200 mg/kg if under 6 years) as a single acute ingestion.
5. *What is the antivenom dose for a child?* The same as for an adult, because the dose neutralises the venom injected, not the body weight.
6. *What is the elemental iron in a 200 mg ferrous sulphate tablet?* About 60–65 mg.
7. *How long do you observe a child after paraffin ingestion?* At least 6 hours; discharge if asymptomatic with normal RR, SpO_2 and chest.

On the ward round

- **What would you do in the first 5 minutes for any poisoned child?** ABC (with self-protection), glucose, seizure control, skin/eye decontamination, then identify the poison and decide on charcoal and antidotes.
- **When would you give activated charcoal?** Within 1–2 hours of a charcoal-binding poison in a child with a safe airway; not for hydrocarbons, corrosives, iron, lithium or alcohol, and not in a drowsy child.
- **How do you know you have given enough atropine?** The chest is clear on auscultation, the heart rate is normal for age, the skin is dry, the pupils are no longer pinpoint, and BP is adequate.
- **This toddler drank paraffin and is well at 6 hours. Can he go home?** Yes, if RR and SpO_2 are normal and there is no cough or chest signs (and a normal chest X-ray if one was done); advise return if cough or fast breathing develops, and counsel on storage.
- **When would you start acetylcysteine before getting a level?** If more than 8 hours have passed since a potentially toxic ingestion, if the dose is ≥150 mg/kg and a level will not be available, for staggered ingestions, or if LFTs are abnormal.
- **What are the indications for antivenom?** Systemic envenoming (neurotoxicity, non-clotting blood or bleeding, shock) or rapidly spreading local swelling (more than half the limb, or beyond the wrist/ankle within hours).

- **A teenager has taken an overdose. What else must you do?** Ask about co-ingestants (assume paracetamol), monitor, and arrange a psychosocial/psychiatric assessment before discharge; involve the family and ensure safety.

Mnemonics

- **"Don't make them vomit" list = "PCB"**: **P**etroleum products, **C**orrosives, **B**rain depressed (drowsy child).
- **Charcoal doesn't bind "PHAILS"**: **P**esticides in hydrocarbon solvents (careful), **H**ydrocarbons, **A**cids/alkalis and **A**lcohols, **I**ron, **L**ithium, **S**olvents.
- **Cholinergic = "SLUDGE + killer Bs"**: Salivation, Lacrimation, Urination, Defecation, GI upset, Emesis; Bradycardia, Bronchorrhoea, Bronchospasm.
- **"Dry the chest"**: the atropine target is a dry chest, not big pupils.
- **NAC "150-50-100 over 1-4-16"**.
- **Iron "20-60"**: <20 mg/kg safe, >60 mg/kg dangerous.
- **Snakebite first aid "RIGHT"**: **R**eassure, **I**mmobilise, **G**et to hospital, **T**ell the health worker the symptoms (no Herbs, no tourniquet).
- **History "5 Ws"**: Who, What, When, How much, Why/how.

OSCE tips

- In a counselling station, start with empathy (parents feel guilty), then give **specific, practical** prevention advice and first-aid dos and don'ts.
- In a history station, ask the **5 Ws** and always ask about **other children** and **intent** (adolescents).
- Show dose calculations: elemental iron mg/kg, paracetamol mg/kg.
- In an examination station, say aloud: pupils, skin moisture, secretions, heart rate, smell, mouth, chest and neurology.

15. References

1. Ministry of Health Uganda. Uganda Clinical Guidelines 2023: section 1.3 Poisoning (general management, organophosphates, paraffin, aspirin, paracetamol, iron, CO, opioids, warfarin, methanol, alcohol) and 1.2.1.1 Snakebites. Kampala: MoH; 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
2. World Health Organization. Pocket Book of Hospital Care for Children, 2nd edition (sections 1.6 Poisoning and 1.9 Common causes of envenoming). Geneva: WHO; 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
3. World Health Organization. Snakebite envenoming (fact sheet). 2023. https://www.who.int/news-room/fact-sheets/detail/snakebite-envenoming
4. Medscape (eMedicine). Organophosphate Toxicity: Treatment & Management. 2024. https://emedicine.medscape.com/article/167726-treatment
5. Medscape (eMedicine). Acetaminophen Toxicity: Treatment & Management. 2024. https://emedicine.medscape.com/article/820200-treatment
6. Medscape (eMedicine). Iron Toxicity: Practice Essentials and Treatment. 2024. https://emedicine.medscape.com/article/815213-overview
7. Medscape (eMedicine). Hydrocarbon Toxicity: Treatment & Management. 2024. https://emedicine.medscape.com/article/821143-treatment
8. Medscape Reference. Pralidoxime (Protopam) dosing, pediatric. 2024. https://reference.medscape.com/drug/protopam-2pam-antidote-pralidoxime-343744
9. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	First priorities in any poisoned child?	**Protect yourself**, then **ABC (ETAT)**, **glucose** (10% dextrose 5 mL/kg if low or unmeasurable in a drowsy child), **control seizures**, decontaminate skin/eyes, then identify the poison and decide on charcoal and antidotes.
2	The '5 Ws' of a poisoning history?	**Who** (age, weight), **What** (substance, strength; bring the container), **When**, **How much** (worst case, mg/kg), **Why/how** (accident, error, self-harm, abuse). Also: other children exposed? first aid given?
3	Should you induce vomiting in a poisoned child?	**Never.** It removes little poison, causes aspiration (especially hydrocarbons) and re-burns the oesophagus (corrosives); salt-water emetics can cause fatal hypernatraemia.
4	Dose and timing of activated charcoal?	**1 g/kg** (UCG 2023: 0.5–1 g/kg; WHO: ≤1 year 1 g/kg, 1–12 years 25–50 g), max 50 g, within **1 hour** (UCG up to 2 h), mixed with 8–10 volumes of water, oral or NG, only if the airway is safe.
5	Poisons NOT bound by activated charcoal?	**Iron, lithium, alcohols (ethanol, methanol), hydrocarbons (paraffin, petrol), acids and alkalis** (also potassium and other metals).
6	Main danger of paraffin (kerosene) ingestion?	**Aspiration pneumonitis** (low viscosity spreads over the lungs and destroys surfactant). CNS depression and seizures can also occur.
7	Management of paraffin ingestion?	ABC; wash skin; **no emesis, no lavage, no charcoal**; oxygen if hypoxic; **observe ≥6 hours**; CXR if symptomatic; no routine antibiotics or steroids; counsel never to store paraffin in soda bottles.
8	Features of organophosphate poisoning?	**SLUDGE** (salivation, lacrimation, urination, defecation, GI upset, emesis) + **killer Bs** (bradycardia, bronchorrhoea, bronchospasm), **pinpoint pupils**, sweating, fasciculations, weakness, seizures, coma; chemical smell.
9	Atropine dosing in organophosphate poisoning (child)?	**0.02 mg/kg IV/IM** (WHO Pocket Book; max 2 mg per dose; UCG 2023: 0.05 mg/kg), **doubling every 5 minutes** until atropinised, then an infusion of 10–20% of the loading total per hour.
10	Endpoints of atropinisation?	**Clear chest** (no bronchorrhoea), **heart rate normal for age**, **dry axillae**, pupils no longer pinpoint, adequate BP. A dry chest is the key target.
11	Pralidoxime dose in children?	**25–50 mg/kg IV over 30 min** (max 2 g), then **10–20 mg/kg/h** infusion (WHO, UCG); most useful within 24–48 h, before enzyme 'ageing'.
12	What is the intermediate syndrome?	Weakness of **neck flexors, proximal limbs, cranial nerves and respiratory muscles** 24–96 h after organophosphate poisoning, after the cholinergic crisis has settled; can cause respiratory failure.
13	Toxic dose of paracetamol in a child?	**≥150 mg/kg** single acute ingestion (UCG 2023: >150 mg/kg, or >200 mg/kg in children under 6 years).
14	IV acetylcysteine regimen?	**150 mg/kg over 1 h → 50 mg/kg over 4 h → 100 mg/kg over 16 h** (21 h; total 300 mg/kg). For <20 kg, volumes of 5% dextrose 3, 7 and 14 mL/kg (WHO). Nearly 100% protective if started within **8 h**.
15	When do you take a paracetamol level and what is the UK treatment line?	At **4 hours** after ingestion or later (earlier levels are uninterpretable). UK treatment line **100 mg/L at 4 h** (US 150 mg/L). If >8 h since a toxic dose, start NAC before the result.

#	Question	Answer
16	Elemental iron in common tablets?	Ferrous sulphate 200 mg ≈ **60–65 mg**; ferrous fumarate 200 mg ≈ 65 mg; ferrous gluconate 300 mg ≈ 35 mg.
17	Iron toxicity thresholds (elemental)?	**<20 mg/kg**: usually only GI upset. **20–60 mg/kg**: possible systemic toxicity. **>60 mg/kg**: serious, possibly lethal (UCG: <40 mg/kg unlikely to cause problems).
18	Five phases of iron poisoning?	(1) **GI** 0.5–6 h: vomiting, haematemesis, pain; (2) **latent** 6–12 h; (3) **shock and acidosis** 12–48 h; (4) **hepatic failure** 2–4 days; (5) **GI stricture** (pyloric) after 2–6 weeks.
19	Deferoxamine dose for iron poisoning?	**15 mg/kg/h IV**, reduced after 4–6 h, max **80 mg/kg/24 h** (WHO; max 6 g/day), usually for **≤24 h**. Indications: shock, acidosis, coma, severe GI symptoms, serum iron >90 µmol/L. Urine turns 'vin rosé'.
20	Why do small children who drink alcohol become comatose and cold?	Ethanol depresses the CNS and **blocks hepatic gluconeogenesis → hypoglycaemia**; peripheral vasodilatation → **hypothermia**. Treat with 10% dextrose 5 mL/kg, dextrose infusion, warmth.
21	Snakebite first aid (UCG 2023)?	Reassure, keep still, lie on the side, remove rings/tight items, **splint the limb**; **no tourniquet, no cutting or sucking, no herbs**, don't chase the snake; get to a facility.
22	How do you do a 20-minute whole-blood clotting test?	Put **2 mL of venous blood in a clean, dry glass tube**, leave undisturbed for **20 minutes**, tip once: blood still liquid = **coagulopathy** (venom-induced). Repeat every 4–6 h (UCG observes at 30 min).
23	Indications for antivenom?	**Systemic envenoming** (neurotoxicity: ptosis, bulbar palsy, respiratory weakness; non-clotting blood/bleeding; shock) or **rapidly spreading local swelling** (hand/foot within 1 h, elbow/knee within 3 h, groin/chest, more than half the limb).
24	Antivenom dose for children?	**The same as for adults** (it neutralises the venom injected, not the body weight). Dilute in 2–3 volumes of saline, IV over 1 h, adrenaline 0.01 mL/kg of 1:1000 IM ready.
25	Caustic ingestion: what not to do?	**No emesis, no neutralisation, no charcoal, no lavage, no blind NG tube.** Airway first; nil by mouth if drooling/stridor; endoscopy within 12–24 h if symptomatic; follow up for oesophageal stricture.
26	Naloxone dose for opioid poisoning in a child?	**0.01 mg/kg IV/IM**, increasing to **0.1 mg/kg** if no response (UCG 2023); repeat every 2–3 min; infusion if needed. Aim to restore breathing.
27	Key prevention advice for families?	Store medicines and chemicals **locked up, out of reach, in original labelled containers**; **never put paraffin or pesticides in soda/water bottles**; don't call medicines sweets; supervise; safe disposal.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy (12 kg) drank from a soda bottle containing paraffin 30 minutes ago. He coughed and gagged at once. Now RR 44, SpO_2 96%, chest clear. His aunt asks whether to make him vomit. What is the MOST appropriate management?

A Induce vomiting with salt water to remove the paraffin
B Gastric lavage with 0.9% saline
C Activated charcoal 12 g by NG tube
D Observe for at least 6 hours with RR and SpO_2 monitoring; no emesis, lavage or charcoal
E Start amoxicillin and prednisolone to prevent pneumonia

2. A 4-year-old (16 kg) from a farming home in Masaka is brought drowsy and drooling with vomiting and diarrhoea. HR 70/min, pupils pinpoint, profuse sweating, widespread crackles and wheeze, fasciculations in the calves. SpO_2 88%. What is the MOST appropriate immediate drug treatment?

A Atropine 0.32 mg IV (0.02 mg/kg), doubling every 5 minutes until the chest is clear and the heart rate normal
B Salbutamol nebuliser and IV ceftriaxone for pneumonia
C Naloxone 0.16 mg IV
D Adrenaline 0.16 mL of 1:1000 IM
E Pralidoxime alone, withholding atropine until oxygen saturation improves

3. The child in the previous question has received 0.3, 0.6, 1.2 and 2.4 mg of atropine. Now: HR 135/min, pupils 4 mm, chest clear, axillae dry, temperature 37.9 °C, agitated. What is the BEST next step? Advanced

A Continue doubling: give 4.8 mg
B Stop atropine completely because he is now tachycardic
C He is atropinised: start an infusion at about 10–20% of the total loading dose per hour and monitor chest, HR and temperature hourly
D Give physostigmine to reverse the atropine
E Give a dose of diazepam and continue doubling

4. A 3-year-old (14 kg) swallowed up to 10 of her pregnant mother's ferrous sulphate 200 mg tablets 2 hours ago. She has vomited three times, once with blood, and has abdominal pain. HR 150, CRT 3 s. Which statement is CORRECT? Advanced

A Give activated charcoal 14 g now
B This is about 15 mg/kg elemental iron and needs no treatment
C Induce vomiting to remove remaining tablets
D Send her home if she is better at 2 hours
E She has taken about 45 mg/kg elemental iron, is symptomatic, and needs IV fluids, an abdominal X-ray and deferoxamine if shock or acidosis develops

5. A 15-year-old girl (45 kg) took 30 paracetamol 500 mg tablets 10 hours ago after a quarrel at home. She is nauseated but otherwise well. A paracetamol level will take 24 hours from a private lab. What is the MOST appropriate management?

A Activated charcoal and discharge if well at 12 hours

B Start IV acetylcysteine immediately (150 mg/kg over 1 h, 50 mg/kg over 4 h, 100 mg/kg over 16 h) and arrange psychosocial assessment

C Wait for the paracetamol level before deciding

D Give vitamin K and observe

E Start acetylcysteine only if her ALT is raised

6. A 2-year-old (11 kg) is found with an open bottle of bleach-like toilet cleaner 20 minutes ago. He is drooling, crying, refuses to swallow and has white patches on his lips and tongue. No stridor. What is the BEST next step?

A Give vinegar or lemon juice to neutralise the alkali

B Keep him nil by mouth, protect the airway, give IV fluids and analgesia, and arrange endoscopy within 12–24 hours

C Pass a nasogastric tube and perform gastric lavage

D Give activated charcoal 11 g

E Induce vomiting with salt water

7. An 8-year-old (24 kg) was bitten on the foot by a large snake in the garden 2 hours ago. Swelling now reaches the knee, blisters are forming, and his 20-minute whole-blood clotting test shows non-clotting blood. Which is the BEST management?

A Apply a tight tourniquet above the knee

B Incise and suck the bite

C Give half the adult antivenom dose because he is a child

D Give oral antibiotics and observe

E Give the full adult dose of polyvalent antivenom IV over 1 hour, with adrenaline drawn up, and repeat the clotting test after 6 hours

8. A 6-year-old (20 kg) is brought unconscious from a wedding where he drank several sachets of waragi. Temperature 35.0 °C, RR 18, SpO_2 95%. What is the MOST important immediate investigation?

A Bedside blood glucose

B Serum ethanol level

C Chest X-ray

D Lumbar puncture

E Serum paracetamol level

9. Which ingestion is activated charcoal MOST likely to benefit, assuming the child is alert and presents within 1 hour?

A Ferrous sulphate tablets

B Paraffin

C Caustic soda

D Carbamazepine tablets

E Ethanol

10. A 4-year-old (16 kg) drank most of a 100 mL bottle of paracetamol syrup 120 mg/5 mL about 90 minutes ago. He is well. No paracetamol level is available. What is the MOST appropriate plan? Advanced

A Reassure and discharge; syrup is never toxic

B Give activated charcoal and start N-acetylcysteine immediately without further thought

C Calculate the dose (≈150 mg/kg); give charcoal if within the time window and the airway is safe, and start IV acetylcysteine with baseline LFTs because a level is unavailable

D Induce vomiting to remove the syrup

E Give vitamin K and repeat INR

11. A 3-year-old is brought 4 hours after paraffin ingestion. She has fever 38.6 °C, RR 56, crackles at the right base and SpO_2 89%. Chest X-ray shows right lower lobe infiltrates. What is the BEST management? Advanced

A Oxygen to SpO_2 ≥94%, admission, maintenance fluids if not drinking, and antibiotics only if fever persists beyond 48 h or she deteriorates

B IV ceftriaxone and dexamethasone immediately

C Activated charcoal by NG tube

D Discharge with oral amoxicillin

E Gastric lavage and salbutamol

12. A 5-year-old (18 kg) bitten on the hand by a cobra 1 hour ago now has bilateral ptosis, drooling and difficulty swallowing; RR 30, SpO_2 93%. Antivenom is being prepared. What is the MOST important additional measure? Advanced

A Give pralidoxime

B Apply a tourniquet to stop further venom spread

C Give IM vitamin K

D Prepare to support ventilation (bag-mask, intubation) and consider an atropine–neostigmine trial

E Give oral fluids to keep him hydrated

13. A mother at Mengo asks how to prevent another poisoning after her toddler drank paraffin. Which advice is MOST important?

A Give milk to the child after any poisoning

B Keep a bottle of salt water ready to induce vomiting

C Never store paraffin or chemicals in soda or water bottles; keep them in labelled containers, locked away and out of reach

D Store paraffin on the floor so it cannot fall

E Teach the child that paraffin tastes bad

14. A 2-year-old (12 kg) is drowsy with slow, shallow breathing (RR 8) and pinpoint pupils after drinking an adult's codeine cough syrup. Skin is dry; chest is clear; HR 100. What should you give?

A Atropine 0.24 mg IV

B Flumazenil IV

C Activated charcoal 12 g by mouth

D Pralidoxime 600 mg IV

E Bag-mask ventilation with oxygen and naloxone 0.12 mg (0.01 mg/kg) IV, repeated and increased to 0.1 mg/kg if needed

15. A 3-year-old (14 kg) took an unknown number of her grandmother's iron tablets. She is completely asymptomatic 6 hours later; abdominal X-ray shows no tablets. What is the MOST appropriate decision?

A Start deferoxamine 15 mg/kg/h for 24 hours to be safe

B Significant toxicity is very unlikely; she can be discharged with advice to return if vomiting, drowsiness or abdominal pain develop, plus prevention counselling

C Give activated charcoal now

D Perform whole-bowel irrigation

E Keep in hospital for 7 days to watch for liver failure

Answers & explanations

1. D · Observe for at least 6 hours with RR and SpO_2 monitoring; no emesis, lavage or charcoal

D is correct: hydrocarbon ingestion is managed **supportively**, with observation for at least 6 hours because pneumonitis can develop over hours. A is wrong: vomiting increases aspiration of hydrocarbons and salt emetics can be fatal. B is wrong: lavage is contraindicated for petroleum products. C is wrong: charcoal does not bind hydrocarbons and provokes vomiting. E is wrong: prophylactic antibiotics and steroids have no proven benefit.

2. A · Atropine 0.32 mg IV (0.02 mg/kg), doubling every 5 minutes until the chest is clear and the heart rate normal

A is correct: this is a **cholinergic crisis** (SLUDGE, bradycardia, bronchorrhoea, miosis, fasciculations) from organophosphate exposure; give **atropine 0.02 mg/kg**, doubling every 5 minutes to a dry chest (UCG uses 0.05 mg/kg), with suction, oxygen and decontamination. B is wrong: the 'pneumonia' is bronchorrhoea, and salbutamol will not dry secretions or treat bradycardia. C is wrong: naloxone is for opioids (pinpoint pupils but no secretions, sweating or fasciculations). D is wrong: this is not anaphylaxis. E is wrong: atropine must not be delayed; pralidoxime is an adjunct.

3. C · He is atropinised: start an infusion at about 10–20% of the total loading dose per hour and monitor chest, HR and temperature hourly

C is correct: a **clear chest, dry axillae, normal-for-age HR and non-pinpoint pupils** mean atropinisation; maintain it with an infusion (total loading 4.5 mg → about 0.45–0.9 mg/h) and titrate. A is wrong: further doubling causes atropine toxicity. B is wrong: stopping risks recurrence of bronchorrhoea; mild tachycardia is expected. D is wrong: physostigmine would worsen the cholinergic poisoning. E is wrong: he is not seizing and further doubling is not indicated.

4. E · She has taken about 45 mg/kg elemental iron, is symptomatic, and needs IV fluids, an abdominal X-ray and deferoxamine if shock or acidosis develops

E is correct: 10 × 60–65 mg = about **650 mg elemental iron ÷ 14 kg ≈ 46 mg/kg**, with GI bleeding and early shock; she needs fluids, X-ray, gas and **deferoxamine 15 mg/kg/h** if shocked or acidotic. A is wrong: charcoal does not bind iron. B is wrong: 15 mg/kg is an underestimate; the elemental dose is about 46 mg/kg and she already has GI bleeding and early shock. C is wrong: never induce vomiting. D is wrong: the latent phase (6–12 h) can mislead; a symptomatic child is admitted.

5. B · Start IV acetylcysteine immediately (150 mg/kg over 1 h, 50 mg/kg over 4 h, 100 mg/kg over 16 h) and arrange psychosocial assessment

B is correct: 15 g ÷ 45 kg = **333 mg/kg**, a toxic dose; she is **>8 hours** post-ingestion, so NAC starts immediately without waiting for the level; intentional overdose needs psychosocial assessment. A is wrong: charcoal is useless at 10 hours and discharge would miss delayed hepatotoxicity. C is wrong: delay beyond 8 hours reduces NAC's protection. D is wrong: vitamin K treats warfarin-type poisoning. E is wrong: ALT may be normal early; waiting loses the protective window.

6. B · Keep him nil by mouth, protect the airway, give IV fluids and analgesia, and arrange endoscopy within 12–24 hours

B is correct: drooling and refusal to swallow suggest significant **caustic injury**; keep nil by mouth, watch the airway, give IV fluids and analgesia, and arrange endoscopy within 12–24 h to grade the burns. A is wrong: neutralisation produces heat and more injury. C is wrong: blind NG tubes and lavage risk perforation. D is wrong: charcoal does not bind corrosives and obscures endoscopy. E is wrong: vomiting re-exposes the oesophagus and risks aspiration.

7. E · Give the full adult dose of polyvalent antivenom IV over 1 hour, with adrenaline drawn up, and repeat the clotting test after 6 hours

E is correct: rapidly spreading swelling and **coagulopathy** are indications for antivenom; children receive the **same dose as adults**, given IV with anaphylaxis treatment ready, and the 20WBCT is repeated after 6 h (redose if still non-clotting). A is wrong: tourniquets worsen local necrosis and cause ischaemia. B is wrong: incision and suction cause infection and bleeding without removing venom. C is wrong: the dose is not weight-based. D is wrong: antibiotics are only for infection and do not treat envenoming.

8. A · Bedside blood glucose

A is correct: alcohol in children causes **hypoglycaemia** (blocked gluconeogenesis), which is lethal and instantly treatable with 10% dextrose 5 mL/kg. B is wrong: an ethanol level rarely changes acute management and is seldom available. C is wrong: a chest X-ray is only needed if aspiration is suspected. D is wrong: LP is considered later if meningitis is suspected and the child is stable. E is wrong: there is no history of paracetamol.

9. D · Carbamazepine tablets

D is correct: carbamazepine is **well adsorbed** by charcoal (and multiple-dose charcoal enhances its elimination). A is wrong: charcoal does not bind iron. B is wrong: paraffin is not bound and charcoal increases aspiration risk. C is wrong: corrosives are not bound and charcoal obscures endoscopy. E is wrong: alcohols are absorbed too quickly and are not adsorbed.

10. C · Calculate the dose (≈150 mg/kg); give charcoal if within the time window and the airway is safe, and start IV acetylcysteine with baseline LFTs because a level is unavailable

C is correct: 100 mL × 24 mg/mL = 2,400 mg ÷ 16 kg = **150 mg/kg**, at the treatment threshold; with no level available, a cautious plan is charcoal (UCG allows up to 2 h; benefit falls after 1 h) and **IV NAC**, with LFTs/INR at baseline and the end of treatment. A is wrong: liquid ingestions can still be toxic at this dose. B is wrong: the decision should follow a dose calculation, not reflex. D is wrong: never induce vomiting. E is wrong: vitamin K is for warfarin-type poisoning.

11. A · Oxygen to SpO_2 ≥94%, admission, maintenance fluids if not drinking, and antibiotics only if fever persists beyond 48 h or she deteriorates

A is correct: early fever and infiltrates after hydrocarbon aspiration are usually **chemical pneumonitis**; treatment is oxygen and supportive care, with antibiotics reserved for persistent fever or evidence of secondary bacterial infection. B is wrong: steroids do not help and routine antibiotics are not recommended. C is wrong: charcoal is contraindicated. D is wrong: a hypoxic child must be admitted. E is wrong: lavage is contraindicated.

12. D · Prepare to support ventilation (bag-mask, intubation) and consider an atropine–neostigmine trial

D is correct: **neurotoxic** envenoming threatens the airway and breathing; be ready to ventilate (by relays of bag-mask if no ventilator) and consider an **anticholinesterase trial** (atropine then neostigmine), which reverses post-synaptic cobra neurotoxicity. A is wrong: pralidoxime is for organophosphates. B is wrong: tourniquets cause ischaemia and are not recommended. C is wrong: vitamin K does not treat neurotoxicity. E is wrong: bulbar palsy makes oral fluids an aspiration risk.

13. C · Never store paraffin or chemicals in soda or water bottles; keep them in labelled containers, locked away and out of reach

C is correct: storing paraffin in **drink bottles** within reach is the main cause of childhood paraffin poisoning in Uganda; safe, labelled, locked storage prevents it. A is wrong: milk can provoke vomiting and does not neutralise poisons (WHO advises water/milk sips only for corrosives). B is wrong: salt-water emesis is dangerous. D is wrong: floor storage puts it within a toddler's reach. E is wrong: toddlers do not reliably learn from warnings.

14. E · Bag-mask ventilation with oxygen and naloxone 0.12 mg (0.01 mg/kg) IV, repeated and increased to 0.1 mg/kg if needed

E is correct: **opioid toxidrome** (coma, respiratory depression, pinpoint pupils, no secretions) needs ventilation and **naloxone 0.01 mg/kg**, increased to 0.1 mg/kg if no response (UCG 2023). A is wrong: atropine treats cholinergic poisoning; this child is dry with a clear chest. B is wrong: flumazenil reverses benzodiazepines and may provoke seizures. C is wrong: charcoal in a drowsy child without airway protection risks aspiration. D is wrong: pralidoxime is for organophosphates.

15. B · Significant toxicity is very unlikely; she can be discharged with advice to return if vomiting, drowsiness or abdominal pain develop, plus prevention counselling

B is correct: GI symptoms of iron toxicity appear within 6 hours; a child **asymptomatic at 6 hours** with no tablets on X-ray is very unlikely to develop serious toxicity (WHO). A is wrong: deferoxamine has side-effects and is reserved for clinical toxicity. C is wrong: charcoal does not bind iron. D is wrong: there are no tablets to clear. E is wrong: prolonged admission is unnecessary for an asymptomatic child.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The farmer's son with 'pneumonia'

A 5-year-old boy (18 kg) from Luweero is brought to a HC IV at 18:00 with vomiting, diarrhoea and difficulty breathing since the afternoon. His father had been spraying tomatoes and left the sprayer and an unlabelled bottle in the house. The clinician at the gate thinks he has pneumonia.

1. What would you look for to confirm or refute pneumonia vs poisoning?

> Look for the **cholinergic toxidrome**: **pinpoint pupils**, **bradycardia**, excessive salivation and lacrimation, sweating, bronchorrhoea (wet crackles), wheeze, fasciculations, weakness, chemical smell, incontinence. Pneumonia gives fever and tachycardia with normal pupils. He has HR 64, pupils 1 mm, drooling, sweating, crackles everywhere, SpO_2 88%.

2. What are your immediate actions?

> Gloves and apron; **remove all clothes**, wash skin and hair with soap and water; suction secretions; **oxygen**; IV access; check glucose; give **atropine 0.02 mg/kg = 0.36 mg IV** (UCG: 0.05 mg/kg = 0.9 mg), then **double every 5 minutes** (0.72, 1.44, 2.88 mg…) until the chest is clear, HR normal and axillae dry.

3. After four doses his chest is clear, HR 128, axillae dry, pupils 3 mm. What next?

> He is **atropinised**: start an infusion at 10–20% of the cumulative dose per hour (cumulative ~5.4 mg → about 0.5–1 mg/h), reassess chest, HR, pupils, temperature, bowel sounds and bladder hourly. Give **pralidoxime 25–50 mg/kg (450–900 mg) IV over 30 min**, then 10–20 mg/kg/h if available, because he has fasciculations. Charcoal only if within 1–2 h of ingestion and airway safe.

4. What complications will you watch for over the next 4 days?

> **Respiratory failure** (secretions, weakness), aspiration pneumonia, **atropine toxicity** (fever, delirium, ileus, retention), seizures, and the **intermediate syndrome** at 24–96 h (neck-flexor, proximal and respiratory muscle weakness); later delayed neuropathy. Refer early if ventilation may be needed.

5. How do you counsel the family before discharge?

> Explain what happened without blame; store pesticides **locked away in labelled original containers**, never in drink bottles or the house; wash sprayer equipment outside; keep children away during spraying; change and wash clothes after spraying; return if weakness or breathing difficulty appears.

Take-away: A child with secretions, small pupils and a slow heart is poisoned until proven otherwise: atropine to a dry chest, decontaminate, and watch for the intermediate syndrome.

Case 2: A teenager's overdose

A 14-year-old girl (42 kg) is brought to Mengo by her aunt at 22:00. She admits to taking 'a lot' of paracetamol 500 mg tablets at about 13:00 after being sent away from school for fees. She vomited twice. Pulse 96, BP 108/66, alert, mildly tender in the right upper quadrant.

1. Estimate the risk.

> Assume the worst: the aunt finds an empty strip of 20 and a half-empty tin of 100 → up to 70 tablets = **35 g ÷ 42 kg ≈ 830 mg/kg**, far above **150 mg/kg**. She is **9 hours** post-ingestion, so liver injury is likely without treatment. Ask about co-ingestants (ARVs, antimalarials, alcohol, herbal).

2. What do you do now, before any results?

> Start **IV acetylcysteine** immediately: **150 mg/kg (6.3 g) over 1 h**, then **50 mg/kg (2.1 g) over 4 h**, then **100 mg/kg (4.2 g) over 16 h** in 5% dextrose. Bloods: paracetamol level (if obtainable), ALT/AST, INR, creatinine, glucose, electrolytes, FBC, pregnancy test. Antiemetic. **Charcoal is too late.**

3. Thirty minutes into the first bag she develops itching, flushing and mild wheeze. What do you do?

> An **anaphylactoid reaction** to NAC (rate-related histamine release). **Pause** the infusion, give an antihistamine (chlorphenamine 10 mg IV) and salbutamol for wheeze, then **restart at a slower rate** once settled. Adrenaline only if true anaphylaxis with shock or airway compromise.

4. At 21 hours, ALT is 850 IU/L and INR 1.8. What next?

> **Continue NAC** at 100 mg/kg/16 h (6.25 mg/kg/h) until ALT is falling and INR <1.3 (or improving); monitor glucose, INR, creatinine, acid–base, consciousness. Refer to Mulago/ICU if encephalopathy, INR >3, acidosis, hypoglycaemia or AKI (King's College criteria; liver transplant is not available in Uganda, so supportive care is everything).

5. What must happen before she goes home?

> A **psychosocial/psychiatric assessment** (suicide risk, depression, abuse, school and family stress); safety planning with the family (remove medicines from the home); involve a counsellor or social worker and the school; follow-up appointment; LFTs normalising.

> **Take-away:** In paracetamol overdose, calculate mg/kg on the worst case; if more than 8 hours have passed, start NAC first and ask questions later; every intentional overdose needs a psychosocial assessment.

Case 3: Snakebite in the evening

A 9-year-old boy (26 kg) was bitten on the right foot while walking home at dusk in Kayunga. His family tied a rope tightly around his thigh, cut the bite with a razor and applied herbs. He arrives at the RRH 3 hours later. The foot and leg are swollen to above the knee, with blisters; he has bleeding gums.

1. What are your first actions?

> ABC; reassure; **remove the tourniquet carefully** (be ready for a surge of venom effects: monitor and have antivenom and adrenaline ready); clean the wound; splint and elevate slightly; IV access in an **unbitten limb**; analgesia with paracetamol (no NSAIDs); tetanus prophylaxis; **20WBCT**.

2. The 20-minute clotting test shows liquid blood. What is the likely snake and type of venom?

> Extensive painful swelling with blistering plus **coagulopathy** suggests a **viper** (puff adder or Gaboon viper: cytotoxic with haemotoxic effects); boomslang causes bleeding with less swelling. Neurotoxic snakes (cobras, mambas) cause ptosis and paralysis.

3. Does he need antivenom, and how will you give it?

> Yes: **coagulopathy with bleeding** and swelling beyond the knee. **Polyvalent antivenom at the full adult dose** (per insert), diluted in 2–3 volumes of 0.9% saline, IV over 1 h, slowly for the first 10–15 minutes. **Adrenaline 1:1000 0.26 mL IM** (0.01 mL/kg) drawn up with chlorphenamine and hydrocortisone at hand.

4. Twenty minutes into the infusion he develops urticaria and wheeze. What do you do?

> **Stop the infusion**, give **IM adrenaline 0.26 mL**, oxygen, salbutamol, chlorphenamine; once stable, **restart antivenom slowly**: the envenoming is more dangerous than the reaction.

5. How do you monitor and when do you repeat antivenom?

> Repeat **20WBCT 6 hours** after the antivenom dose; if still non-clotting, **repeat antivenom**. Monitor swelling (mark the edge with times), compartment pressures/pulses, Hb, urine output and colour, creatinine (AKI), bleeding. Surgical review for necrosis (debride once demarcated); fasciotomy only for proven compartment syndrome; antibiotics because the wound was cut and herbs applied.

6. What prevention advice do you give?

> Wear shoes and carry a torch at night, clear grass around the home, sleep on a raised bed under a net; first aid is **immobilise and go to hospital**, never tourniquets, cutting, sucking or herbs.

Take-away: Traditional first aid often worsens snakebite; immobilise, test clotting, and give the full adult dose of antivenom for systemic or rapidly spreading envenoming, with adrenaline ready.

References

1. Uganda Clinical Guidelines 2023: section 1.3 Poisoning (general management, organophosphates, paraffin, aspirin, paracetamol, iron, CO, opioids, warfarin, methanol, alcohol) and 1.2.1.1 Snakebites. *Ministry of Health Uganda*, 2023. https://library.health.go.ug/uganda-clinical-guidelines-2023
2. Pocket book of hospital care for children, 2nd edition (sections 1.6 Poisoning and 1.9 Common causes of envenoming). *World Health Organization*, 2013. https://www.who.int/publications/i/item/978-92-4-154837-3
3. Snakebite envenoming (fact sheet). *World Health Organization*, 2023. https://www.who.int/news-room/fact-sheets/detail/snakebite-envenoming
4. Organophosphate Toxicity: Treatment & Management (Medscape). *Medscape / eMedicine*, 2024. https://emedicine.medscape.com/article/167726-treatment
5. Acetaminophen Toxicity: Treatment & Management (Medscape). *Medscape / eMedicine*, 2024. https://emedicine.medscape.com/article/820200-treatment
6. Iron Toxicity: Practice Essentials and Treatment (Medscape). *Medscape / eMedicine*, 2024. https://emedicine.medscape.com/article/815213-overview
7. Hydrocarbon Toxicity: Treatment & Management (Medscape). *Medscape / eMedicine*, 2024. https://emedicine.medscape.com/article/821143-treatment
8. Pralidoxime (Protopam) dosing, pediatric (Medscape drug reference). *Medscape Reference*, 2024. https://reference.medscape.com/drug/protopam-2pam-antidote-pralidoxime-343744

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.